8th Oct, 2025

D.T. Fourier Series

C.T.

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \phi_k(t)$$

$$= \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

where ω_0 = fundamental frequency of $x(t)$

a_k's were found for all $k = -\infty$ to $+\infty$.

Consider a discrete time periodic signal,

$$x[n] = x[n+N] \quad \forall n$$

for some smaller than the N that this holds true.

N = fundamental time period of the signal $x[n]$.

$$\phi_k[n] = e^{j k \frac{2\pi}{N} n}$$

$$\phi_0[n] = 1 \quad \forall n$$

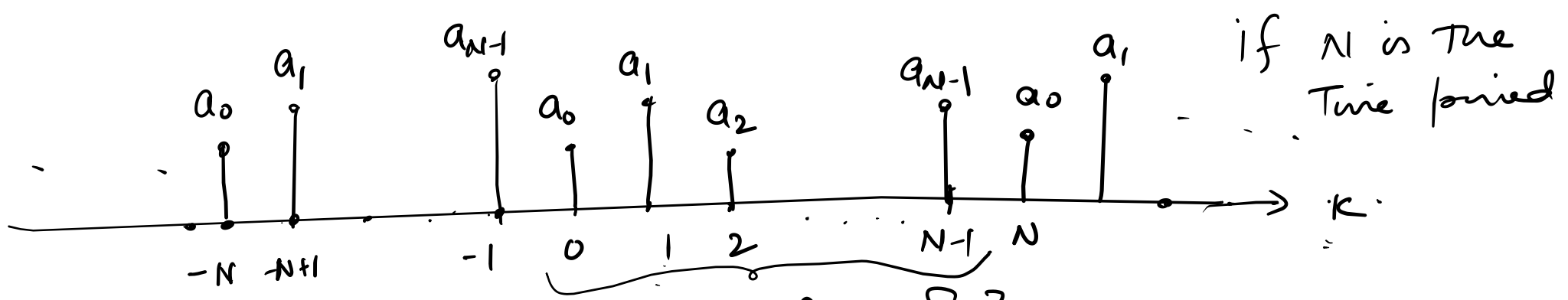
$$\phi_N[n] = e^{j 2\pi n} = \left(e^{j2\pi}\right)^n = 1 \quad \forall n$$

$$\phi_1[n] = e^{j\frac{2\pi}{N}n} \; ; \quad \phi_{N+1}[n] = e^{j(N+1)\frac{2\pi}{N}n} = e^{j2\pi n} \cdot e^{j\frac{2\pi}{N}n}$$

$$= \phi_1[n]$$

$$\phi_0[n] = \phi_N[n] = \phi_{2N}[n] = \phi_{-N}[n] = \cdots = 1$$

$$= \phi_1[n] = \phi_{2N+1}[n] = \phi_{-N+1}[n] = \cdots$$



if N is the Time period

Discrete time periodic signal $x[n]$

the F.S. coefficients also repeat over a period N.

Synthesis = F.S. representation equation

$$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\frac{2\pi}{N}n} = \sum_{k=\langle N \rangle} a_k \phi_k[n]$$

$\langle N \rangle \equiv$ any contiguous N frequency pts.

$\{\phi_k[n]\}_{k=\langle N \rangle} \equiv$ basis functions.

$$x[n] = \sum_{k=<N>} a_k e^{jk\frac{2\pi}{N}n}$$

Multiply both sides with $e^{-jr\frac{2\pi}{N}n}$

$$\sum_{n=<N>} x[n]\, e^{-jr\frac{2\pi}{N}n} = \sum_{n=<N>}\sum_{k=<N>} a_k e^{j(k-r)\frac{2\pi}{N}n} \quad \text{—— (1)}$$

$$= \sum_{k=<N>} a_k \sum_{n=<N>} e^{j(k-r)\frac{2\pi}{N}n}$$

$$\sum_{n=<N>} e^{j(k-r)\frac{2\pi}{N}n} = \begin{cases} 0 & \text{if } k \neq r \\ \sum_{n=<N>} e^{j0} & k = r \end{cases}$$

$$= \begin{cases} 0 & \text{if } k \neq r \\ N & \text{if } k = r \end{cases} \quad \text{—— (2)}$$

Substituting ② in ①, we obtain

$$\sum_{n=\langle N\rangle} x[n]\, e^{-jr\frac{2\pi}{N}n} = N.a_r$$

$$\text{or} \quad a_r = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]\, e^{-jr\frac{2\pi}{N}n}$$

$$\text{or} \quad a_k = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]\, e^{-jk\frac{2\pi}{N}n} \equiv \text{Analysis eqn.}$$

$$x[n] = \sum_{k=\langle N\rangle} a_k\, e^{jk\frac{2\pi}{N}n} \equiv \text{Synthesis eqns.}$$

F.S. coefficients or The Spectral coefficients $\equiv$ form the orthogonal basis of the vector space.

$$\{\phi_k[n]\}_{k=\langle N\rangle}$$

$$a_k = \langle x[n], \phi_k[n]\rangle_N = \frac{1}{N}\sum_{n=\langle N\rangle} x[n]\,\phi_k^*[n]$$

Ex

$x[n] = \sin \omega_0 n$ $\qquad \omega_0 = \frac{2\pi}{N}$

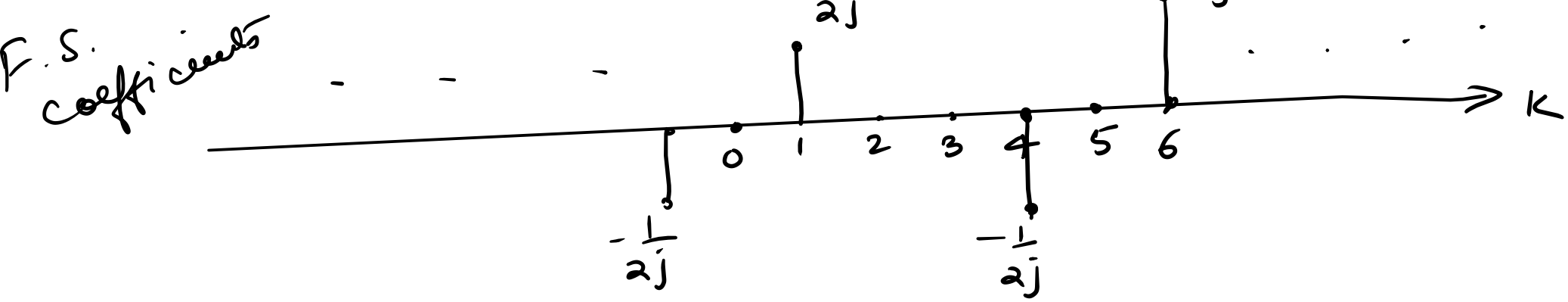
Let $N = 5$

$$= \frac{e^{j\omega_0 n} - e^{-j\omega_0 n}}{2j}$$

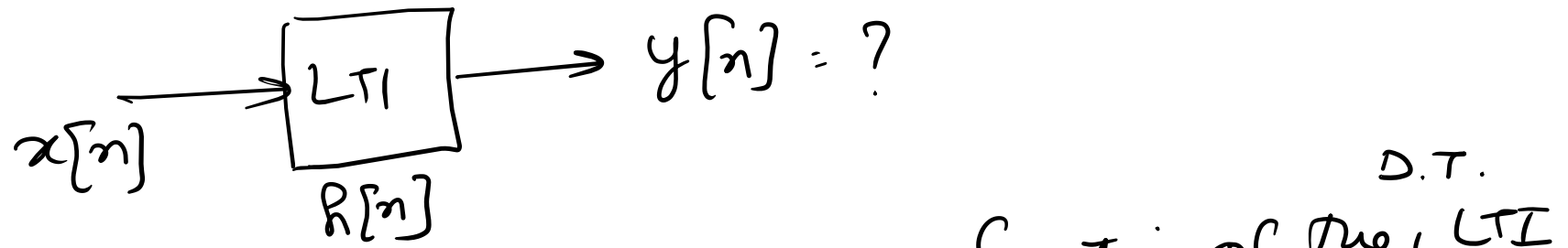
F.S. coefficients:

$a_0 = 0$, $a_1 = \frac{1}{2j}$, $\cdots$ $a_{N-1} = \frac{-1}{2j}$

$a_N = 0$, $a_{N+1} = \frac{1}{2j}$, $\cdots$



Consider an LTI system in the discrete-time $h[n]$



D.T. LTI system.

Say $x[n] = z^n \equiv$ acts as eigenfunction of the

$$y[n] = x[n] * h[n]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

$$= \sum_{k=-\infty}^{\infty} h[k]\, z^{n-k} = \sum_{k=-\infty}^{\infty} h[k]\, z^n \cdot z^{-k}$$

$$= z^n \underbrace{\sum_{k=-\infty}^{\infty} h[k]\, z^{-k}}_{\text{Define } H(z) \triangleq \sum_{n=-\infty}^{\infty} h[n] z^{-n}} = z^n H(z)$$

Let $z^n = \left(e^{j\omega_0}\right)^n$ $\quad z = e^{j\omega_0}$

if $x[n] = z^n = e^{j\omega_0 n}$

$y[n] = H\left(e^{j\omega_0}\right) \cdot e^{j\omega_0 n}$

if $x[n]$ is periodic & $e^{jk\omega_0 n}$

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n}$$

then, $y[n] = \sum_{k=<N>} a_k H\left(e^{jk\omega_0}\right) e^{jk\omega_0 n}$

$y[n]$ is also periodic with F.S. coefficients

$$\boxed{b_k = a_k H\left(e^{jk\omega_0}\right)}$$

Ex $h[n] = \alpha^n u[n] \qquad -1 < \alpha < 1$

& $x[n] = \cos\left(\frac{2\pi}{N} n\right)$ ≡ periodic with period N. $\omega_0 = \frac{2\pi}{N}$

Find $y[n] = ?$

$$= \sum_{k=<N>} a_k H\left(e^{jk\omega_0}\right) e^{jk\omega_0 n}$$

$x[n]$ has only two non-zero coefficients

$a_1 = \frac{1}{2}$ & $a_{-1} = \frac{1}{2}$

$$y[n] = \frac{1}{2} H\left(e^{j\omega_0}\right) e^{j\omega_0 n} + \frac{1}{2} H\left(e^{-j\omega_0}\right) e^{-j\omega_0 n}$$

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n} = \sum_{n=-\infty}^{\infty} \alpha^n u[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} \alpha^n z^{-n} \qquad -1 < \alpha < 1$$

$$= \sum_{n=0}^{\infty} (\alpha z^{-1})^n = \frac{1}{1 - \alpha z^{-1}}$$

$$H(e^{j\omega_0})\Big|_{z=e^{j\omega_0}}$$

$$H(e^{-j\omega_0})\Big|_{z=e^{-j\omega_0}}$$